

Dhareshwar Vidya Va Krida Pratisthan's
Dhareshwar Shikshansastra Mahavidyalaya

Dhayari Pune - 41

B.Ed. First Year

B.ED. 104-09: PEDAGOGY OF THE SCHOOL SUBJECT

(Mathematics)

Name of the Student: Mr./Mrs. Rohini Ramesh Patil Roll No.: 04

Practical No. 1: Content Analysis of Mathematics Unit

EVALUATION SCHEME

Rating Scale 1) Unsatisfactory 2) Average 3) Satisfactory 4) Good 5) Excellent

No	Criteria	1	2	3	4	5
1	Clarity and Relevance of Objectives उद्दिष्टांची स्पष्टता व सुसंगती				✓	
2	Selection of Unit संकल्पनात्मक रचना व तार्किक अनुक्रम				✓	
3	Conceptual Structure and Logical Sequencing विषयवस्तूची अचूकता व गणितीय काटेकोरपणा				✓	
4	Content Accuracy and Mathematical Rigor विषयवस्तूची अचूकता व गणितीय काटेकोरपणा				✓	
5	Examples and Illustrations उदाहरणे व स्पष्टीकरणे				✓	
6	Exercises and Questions सराव प्रश्न व क्रिया				✓	
7	Pedagogical Approach अध्यापन दृष्टिकोन				✓	
8	Language and Presentation भाषा व सादरीकरण				✓	
9	Critical Analysis and Reflection समीक्षात्मक विश्लेषण व आत्मपरिक्षण				✓	
10	Educational Implications and Overall Impression शैक्षणिक परिणाम व एकूण प्रभाव				✓	
TOTAL MARK - 50		40				
MARKS OUT OF - 25		20				

Signature of Professor In-charge

SELECTION OF UNIT

The unit Linear Equations in one variable is well-selected and suitable for std. VIII students.

It build upon student's prior knowledge of basic algebraic expressions learned in earlier classes.

The Topic acts as a foundation for advanced algebraic concepts taught in higher classes like simultaneous equations, quadratic equations and functions.

The unit is :

- Age-appropriate
- curriculum-oriented
- Useful for future mathematical learning

Thus, the Selection of this unit is justified and educationally sound.

* Pedagogical Approach -

The pedagogical approach followed in unit reflects a construction and learner-centered methodology. Concepts are introduced through familiar situations and gradually formalized into algebraic representations.

Students are encouraged to think logically, translate verbal statements into equations and verify their solutions. The use of guided examples, classroom discussion, and problem-solving activities promotes active learning. The unit supports independent thinking and systematic reasoning.

The pedagogical approach promotes analytical and reasoning skills rather than rote learning. Students are encouraged to understand the logic behind each operation and justify the steps taken while solving equations. This prepares them for advanced algebraic concepts and develops mathematical maturity.

The teaching approach used in this unit fosters conceptual clarity, learner engagement and independent problem-solving skills.

Educational Implications and

Overall Impression

The unit linear equation in one variable has significant educational value as it lays a strong foundation for algebraic thinking among class 8 students. It introduces learners to the systematic use of symbols, variables and equations, which are essential for advanced mathematical learning. The concepts developed in this unit serve as prerequisites for future topics such as simultaneous equations, polynomials, linear inequalities and co-ordinate geometry.

The unit also promotes the application of mathematics in real life situation by solving word problems based on daily experiences. Students understand the practical relevance of algebra. This helps reduce fear of abstract mathematics and builds a positive attitude towards the subject. The habit of verifying solutions further develops accuracy, self correction skills and confidence.

Overall the unit is also beneficial for students of varying abilities. Clear explanation and simple problems support slow learners while applications - based and multi-step questions challenges average and advanced

धारेश्वर विद्या व क्रीडा प्रतिष्ठान संचलित

धारेश्वर शिक्षणशास्त्र महाविद्यालय

इंद्रप्रस्थनगर, धायरी, पुणे - ४११ ०४१

अंतर्गत सैध्दांतिक मूल्यमापन

प्रथम वर्ष/द्वितीय वर्ष : सन २०२५ - २०२६

प्रशिक्षणार्थीचे नांव : Rohini Ramesh patil रोल नं.: ०५
विषय सांकेतांक व विषय 101-09 Mathematics
दिनांक : ०९/०१/२०२६ पर्यवेक्षकाची सही [Signature] ०९/०१/२०२६

प्रश्न क्र.	१	२	३	४	५	६	७	८	९	१०	एकूण
गुण	०५	०७	०८	०६	—	—	—	—	—	—	२६ ३५

Q. 1.

1. ~~b~~ → logical thinking

2. ~~c~~ → $5+3+3+4$

3. ~~c~~ → Heuristic method

4. ~~c~~ → Formative assessment

5. ~~c~~ → learning by doing

Q. 2.

Q.2 A

1) Meaning of mathematics -

mathematics is the study of numbers, quantities, shape, patterns and relationship. It deals with logical reasoning and problem solving. It provides both tools for calculation, measurement and analysis in everyday life.

2) Nature of Mathematics -

- Abstract and logical: mathematics deals with abstract concepts and logical reasoning.

- Exact science: it gives precise and accurate results.

- Universal: mathematics principles are valid everyday, irrespective of culture or region. cumulative knowledge in mathematics is built step by step.

3) Scope of mathematics -

mathematics has wide application in various fields:

- Education - Helps in developing problem solving and analytical skills.

- Science and Technology: essential for physics, chemistry, engineering

and Computer science -

- Daily life: used in shopping, banking, budgeting and planning.

- Research and statistics -

mathematics provides tools for data analysis and research studies.

4) Importance of mathematics -

- Develops logical thinking and reasoning ability.

- Helps in decision making and problem solving.

- provides scientific approach to life.

- Useful in everyday activities like measurement, calculation and estimation.

plays a vital role in career opportunities in engineering, technology, research and management.

mathematics is an essential subject that helps in the overall development of students mind, reasoning and analytical skills.